

- When the distance traveled by an object is very large compared to its dimensions, the object can be modeled by a **particle**.
- A group of particles is called **system of particles**.
- **Deformable system**: the distance between its points varies.
- **Non-deformable (or rigid) system**: the distance between any two points remains fixed.
- Suppose a **system (S)** consists of several particles, $P_1, P_2, P_3, \dots, P_n$, of respective masses $m_1, m_2, m_3, \dots, m_n$. Then the **position of the center of mass "G"** of the system with respect to an origin "O" is given by :

$$x_G = \frac{m_1 \cdot x_1 + m_2 \cdot x_2 + m_3 \cdot x_3 + \dots + m_n \cdot x_n}{m_1 + m_2 + m_3 + \dots + m_n} \quad \text{and} \quad y_G = \frac{m_1 \cdot y_1 + m_2 \cdot y_2 + m_3 \cdot y_3 + \dots + m_n \cdot y_n}{m_1 + m_2 + m_3 + \dots + m_n}$$

- **Internal forces**: result from interactions between different parts of the same system.
- **External forces**: result from interactions between the system or part of the system and the environment.
- The point of application of the weight of a system is called the **Center of gravity** of the system. In most cases, the center of gravity coincides with the center of mass of the system.
- To understand analytically **the movement of the center of mass** let us take again the definition of the position vector of the center of mass:

$$\vec{r}_G = \frac{m_1 \cdot \vec{r}_1 + m_2 \cdot \vec{r}_2 + m_3 \cdot \vec{r}_3 + \dots + m_n \cdot \vec{r}_n}{m_1 + m_2 + m_3 + \dots + m_n}, \quad \text{or}$$

$$(m_1 + m_2 + m_3 + \dots + m_n) \cdot \vec{r}_G = m_1 \cdot \vec{r}_1 + m_2 \cdot \vec{r}_2 + m_3 \cdot \vec{r}_3 + \dots + m_n \cdot \vec{r}_n$$

By derivation relative to time:

$$(m_1 + m_2 + m_3 + \dots + m_n) \cdot \frac{d\vec{r}_G}{dt} = m_1 \cdot \frac{d\vec{r}_1}{dt} + m_2 \cdot \frac{d\vec{r}_2}{dt} + m_3 \cdot \frac{d\vec{r}_3}{dt} + \dots + m_n \cdot \frac{d\vec{r}_n}{dt}$$

$$\Rightarrow M \cdot \vec{V}_G = m_1 \cdot \vec{V}_1 + m_2 \cdot \vec{V}_2 + m_3 \cdot \vec{V}_3 + \dots + m_n \cdot \vec{V}_n$$

Deriving the last equation relative to time, we get:

$$M \cdot \frac{d\vec{V}_G}{dt} = m_1 \cdot \frac{d\vec{V}_1}{dt} + m_2 \cdot \frac{d\vec{V}_2}{dt} + m_3 \cdot \frac{d\vec{V}_3}{dt} + \dots + m_n \cdot \frac{d\vec{V}_n}{dt} \Rightarrow M \cdot \vec{a}_G = \underbrace{m_1 \cdot \vec{a}_1}_{\vec{f}_1} + \underbrace{m_2 \cdot \vec{a}_2}_{\vec{f}_2} + \underbrace{m_3 \cdot \vec{a}_3}_{\vec{f}_3} + \dots + \underbrace{m_n \cdot \vec{a}_n}_{\vec{f}_n}$$

$$\Rightarrow \boxed{M \cdot \vec{a}_G = \sum \vec{F}_{\text{external/System}}}$$

This relation expresses **the theorem of the center of mass** of a material system: **"the movement of the center of mass of a system is that of a particle, of mass equal to that of the system, and subjected to a force equal to the sum of the external forces applied to the system"**.

- Particular case

If the sum of the external forces is zero ($\sum \vec{F}_{\text{ext}} = \vec{0}$), the center of mass of the system can be either at rest or in uniform rectilinear motion. Indeed :

$$\sum \vec{F}_{\text{ext}} = \vec{0} \Leftrightarrow M \cdot \vec{a}_G = \vec{0} \Rightarrow \vec{a}_G = \vec{0} \Rightarrow \vec{V}_G = \overline{\text{constant}} \Rightarrow \begin{cases} \text{"G" remains at rest if it is initially at rest} \\ \text{or} \\ \text{"G" continues in URM if it is initially moving} \end{cases}$$